Tanish Verma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.54
FIITJEE junior College Miyapur	Class XII (Telangana State Board of Intermediate Education)	2024	97.60%
The Global Edge School	Class X (CBSE)	2022	90.80%

PROJECTS

xv6-RISC-V Kernel Extensions | C | GitHub

- Designed and implemented a **system-call-aware Multi-Level Feedback Queue (MLFQ) scheduler** featuring four priority levels, adaptive demotion based on CPU vs. I/O behavior, periodic priority boosting, and runtime scheduling statistics exported through custom kernel system calls.
- Designed and implemented a **demand-paged virtual memory subsystem** supporting lazy allocation, clock-based replacement, and disk-backed swapping, enabling memory oversubscription while maintaining synchronization across concurrent page-fault and swap paths.
- Built a **RAID-backed storage layer** supporting RAID 0/1/5, SSTF/FCFS disk scheduling, parity-based recovery, and per-process disk statistics, enabling fault-tolerant swap management under simulated disk failures.

Telegram To-Do Bot | Python | SQLite | GitHub

- Developed an **asynchronous, multi-user Telegram bot** using python-telegram-bot and aiosqlite, supporting persistent task management with user-isolated data storage.
- Designed an **event-driven architecture** with inline keyboard interactions, confirmation workflows, and non-blocking database operations to provide a responsive user experience.
- Containerized the application with **Docker** and configured both polling and webhook deployment modes using environment-based configuration for portable deployment.

Numerical BVP Solver | Python | Team of 4 | GitHub

- Developed a modular Python framework for solving second-order boundary value problems using the **Shooting Method** (RK2/RK4/ABM2/ABM4) and **Finite Difference Method (FDM)** with **Newton iteration**, collaborating in a four-member team.
- Implemented **RK4 integration** and performed **grid-refinement convergence analysis** to validate theoretical convergence rates; derived a **Residual-Jacobian formulation** to resolve numerical instability caused by an artificial singularity.
- **Architected** the repository structure and **configuration-driven CLI**, integrating independently developed solver modules into a unified, extensible numerical framework.

Molecular Dynamics Simulation Framework | Python | Team of 4 | GitHub

- Architected a **modular simulation framework** centered around a reusable MDSimulation engine, designing the project structure, interactive CLI, configuration system, and HDF5-based persistence pipeline to support reproducible computational experiments.
- Integrated lattice generation, neighbor-list construction, velocity-Verlet integration, diagnostics, and visualization into a unified execution pipeline capable of offline analysis from serialized simulation outputs.
- Designed a scalable batch-execution workflow that performed **4000+ independent simulation jobs** on a SLURM-managed HPC cluster, enabling automated generation and post-processing of complete temperature-density phase maps.

SKILLS

- **Languages:** Python, C, C++
- **Libraries & Frameworks**: NumPy, Pandas, SciPy, Matplotlib, Streamlit

- **Core CS**: Data Structures & Algorithms, Operating Systems.
- **Developer Tools**: Git, GitHub, Linux, Jupyter Notebook.

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Academic Excellence Award** at IIT Hyderabad for securing the **highest CGPA** in the Jul-Dec 2024 semester; maintained the highest CGPA **in Computational Engineering** for **three consecutive semesters**.
- Ranked in the **top 8% nationwide** among **250,000+ participants** in the **AlgoUniversity Tech Fellowship (ATF) 2025 - Stage 1**.
- Earned **A+ grades** in **Operating Systems I & II**, **Electronic Structure and Atomistic Modeling of Materials**, and **Differential Equations**.

RELEVANT COURSES

- **Computer Science**: Data Structures and Applications, Operating Systems, Programming in C, Digital Circuits, Introduction to Hardware Description Language.
- **Mathematics**: Calculus, Linear Algebra, Probability, Statistics, Differential Equations.
- **Scientific Computing**: Fundamentals of Scientific Computing, Computer-Aided Numerical Methods.

POSITIONS OF RESPONSIBILITY

- **Internship Coordinator, OCS 2026-27:** Coordinated internship outreach for the Computational Engineering batch, facilitating communication between students, the Office of Career Services, and recruiting companies throughout the internship season.

MISCELLANEOUS

- Solved **400+** Data Structures & Algorithms problems using the Take U Forward DSA Sheet.